AUTONOMOUS MAZE SOLVER BOT — USER MANUAL & SYSTEM DESCRIPTION

1. Introduction

This document describes the working, control logic, and user interaction of the Autonomous Maze Solver Bot. The system is designed to explore an unknown maze, detect exits, compute the optimal path, and execute a fast run.

The robot uses real-time sensing, encoder-based motion tracking, and hybrid pathfinding algorithms to navigate reliably in complex maze environments.

2. System Overview

The system consists of:

- ESP32-based controller
- Front ToF sensor (VL53L5CX)
- Left & Right ToF sensors (VL53L0X)
- Motor driver with encoder feedback
- Bluetooth interface for telemetry
- Push button for control and mode selection

The bot operates in three phases:

1. Exploration
2. Path Optimization
3. Fast Run

An additional **manual wall-follow mode** is available.

3. Startup Procedure

After powering ON:

- System waits in idle state
- Start can be triggered using:
    - Push button (GPIO 0)
    - Bluetooth command ("R")

4. Push Button Functions (Exact Usage)

The push button is used to control modes and actions. The behavior depends on how many times it is pressed or how long it is held.

4.1 Mode Selection (Before Start)

- **Press 1 time → Prefer Front Mode**
  Robot prefers moving straight.
- **Press 2 times → Prefer Right Mode**
  Robot prefers right turns.
- **Press 3 times → Prefer Left Mode**
  Robot prefers left turns.

The modes cycle continuously:
Front → Right → Left → Front ...

Audio feedback:

- 1 beep → Front
- 2 beeps → Right
- 3 beeps → Left

4.2 Long Press (≥ 2 seconds)

- **Press and hold for 2 seconds → Start Wall Following Mode**

Behavior:

- A long beep confirms activation
- Robot immediately starts wall following
- No mapping or maze solving happens in this mode

4.3 While Wall Following is Running

- **Press button once → Switch wall side**

Behavior:

- Right wall → switches to Left wall
- Left wall → switches to Right wall

A short beep confirms the change.

4.4 Goal Selection (After Exploration)

- **Press 1 time → Select Goal 1**
- **Press 2 times → Select Goal 2**
- **Press 3 times → Select Goal 3**
- **Press 4 times → Select Goal 4**

Rules:

- All presses must happen within ~600 ms
- After timeout, selection is locked

4.5 Fast Run Start

- Wait 1 second after path computation
- **Press button once → Start Fast Run**

5. Working Principle

5.1 Initial Behaviour: Wall Following

Wall following is activated only by long press.

The robot:

- Maintains constant distance from wall using PID
- Uses front sensor for collision avoidance
- Applies small angular corrections (≈20° turns)
- Continuously scans for exit condition

5.2 Exit from Wall Following (STRICT CONDITIONS)

The robot exits wall-following mode only when **all conditions below are satisfied in sequence**:

Stage 1 — Front Opening Detection

- Front ToF center pixels detect distance **≥ 1000 mm**
- Data must be valid (target status ≠ 255)

At this instant:

- Encoder values are stored (trigger point)

Stage 2 — Straight Motion Validation

From the trigger point:

- Robot must travel **≥ 300 encoder ticks (average of encA and encB)**
- Encoder difference must satisfy:

$$
|encA - encB| < 95
$$

Meaning:

- Robot must move straight (no sharp turn)

If this condition fails:

- Trigger is cancelled
- Process restarts

Stage 3 — Side Opening Detection

After successful Stage 2:

- Robot waits for side opening:

Stage 4 — Centering

After side gap detection:

- Robot drives forward **680 encoder ticks**
- Stops exactly at cell center

Stage 5 — Transition

- 500 ms confirmation beep
- Control exits wall follower
- Main maze-solving algorithm starts

5.3 Phase 1: Exploration

The robot performs full maze exploration using:

- Hybrid Tremaux algorithm
- Flood fill distance mapping

Key Operations

- Detect walls using ToF sensors
- Mark visited cells
- Store up to 4 exits
- Maintain distance map

Exit Detection

Exit is detected when:

- Front > 1000 mm
- Left > 700 mm
- Right > 700 mm

Exploration Stops When:

- All reachable cells are visited
- OR shortest path is proven
- OR no further progress is detected

5.4 Phase 2: Path Optimization

After exploration:

1. User selects goal using button
2. System isolates that goal
3. Computes path

Algorithms used:

- DFS optimization
- Flood fill (fallback)

Final path is stored and transmitted.

5.5 Phase 3: Fast Run

Robot executes optimal path at high speed.

Behavior:

- Uses distance map
- Chooses minimum distance direction
- Minimizes turns

Special Case

If wall-follow was used initially:

- Fast wall-follow runs first
- Then transitions into grid navigation

6. Motion Control

6.1 Centering (PID)

- Uses left/right distance difference
- Maintains center alignment

6.2 Encoder Balance

- Corrects motor mismatch
- Keeps straight motion

6.3 Node Detection

A node is detected when:

- Wall pattern changes
- Sudden distance variation
- Confirmed over multiple readings

Robot then moves fixed ticks to reach center.

6.4 Pin Detection

- Uses front sensor grid
- Detects small obstacles
- Applies lateral shift
- Ensures smooth avoidance

7. Sensors

- Front: Multi-zone ToF (VL53L5CX)
- Side: Single-point ToF (VL53L0X)

Features:

- Invalid data rejection
- Averaging of center pixels
- Dynamic wall compensation

8. Safety Features

- Encoder-based validation
- Motion timeout protection
- PID reset after turns
- Button debounce
- Multi-stage exit verification

9. Complete Flow

1. Power ON
2. Select mode (optional)
3. Start system
4. (Optional) Wall following
5. Exit wall-follow (strict condition)
6. Exploration
7. Goal selection
8. Path optimization
9. Fast run

10. Finish

10. Conclusion

The system integrates sensing, control, and intelligent algorithms to solve complex mazes efficiently. The use of strict conditions, encoder validation, and hybrid algorithms ensures reliable and repeatable performance.

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